

Mikel Egaña Aranguren, Ph.D.

Associate Professor

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Education

2009 **Ph.D.** Computer Science, University of Manchester, UK

2005 **M.Sc.** Bioinformatics, University of Manchester, UK

2003 **B.Sc.** Biology, University of Basque Country, Spain

2002 **Invited student** Evolutionary Ecology at Canterbury Christ Church University College, UK

2002 **Erasmus student** Environmental Biology at Canterbury Christ Church University College, UK

Employment

2024/07/01 - Present **Associate Professor**, Dept. of Computer Languages and Systems, EHU

2020/09/01 - 2024/06/30 **Assistant Professor**, Dept. of Computer Languages and Systems, EHU

2018/12/02 - 2020/08/31 **Linked Data Consultant** Eccenca GmbH
Development of Linked Data solutions for the Enterprise, including project management

2018/10/01 - 2018/12/02 **Bioinformatics technician** Biocruces Bizkaia
Data infrastructure development, Bioinformatics analyses

2016/01/14 - 2018/09/30 **Analyst, Torres Quevedo fellow** Eurohelp Consulting
Design and development of Linked Open Data solutions

2015/04/27 - 2016/01/14 **Analyst** Eurohelp Consulting
Design and development of Linked Open Data solutions

Research positions

- 2014/04/01 - 2015/03/31 **Post-doc researcher (80% FTE)** Genomic Resources Group, EHU
Metagenomics and Life Sciences Semantic Web
- 2011/02/14 - 2014/02/14 **Post-doc researcher, Marie Curie Cofund fellow**
Ontology Engineering Group (Computer Science);
Biological Informatics Group (CBGP), UPM, Spain
Ontology Engineering and Life Sciences Semantic Web
- 2010/12/01 - 2011/02/01 **Researcher** OGO project, UM, Spain
Orthologous Genes Ontology
- 2006/05/01 - 2006/10/1 **Pre-doc researcher, Marie Curie EST fellow**
Computational Biology group, VIB, Belgium
Cell Cycle Ontology and Ontology Design Patterns

Freelance positions

- 2018/10/15 - 2018/12/02 **Ontology Engineering.** Cognizone, Belgium
- 2018/05/01 - 2018/07/31 **Public tender technical writer.** University of Murcia
Project Hercules: Federated Linked Open Data for universities

Research visits

- 2005/09/01 - 2005/10/01 European Bioinformatics Institute (EBI), funded by the Network of Excellence on Semantic Interoperability and Data Mining in Biomedicine (EU)

Personal funding

- 2018-2020 (2 years) **Declined** Bioinformatics Technician. Instituto de Salud Carlos III. CA18/00021
- 2016-2018 (3 years) Torres Quevedo (Spain). 35% of salary at Eurohelp Consulting. PTQ-14-07198
- 2011-2014 (3 years) Marie Curie Cofund (EU). UNITE 246565
- 2006 (5 months) Marie Curie EST (EU). MEST-CT-2004-414632
- 2005 (One payment) EPSRC (UK): Ph.D. fees
- 2005 - 2008 (3 years) University of Manchester (UK): Ph.D. maintenance allowance
- 2002 (5 months) Erasmus (EU)

Participation in projects

Research

- 2025 EuskoPVmap. 20.000,00 EUR.
- 2024 - 2027 BRAINMAP-REHAB (Biocruces)
- 2024 EHUKhi: una plataforma para el desarrollo del potencial fotovoltaico de la EHU (Campus Bizia Lab). 2.225,00 EUR.
- 2021 - 2025 SUPPORT4LHS (Process Mining and Knowledge Representation technologies to Support the Learning Health System). PID2020-113723RB-C22. Agencia Estatal de Investigación, “Proyectos I+D+i 2020” - Modalidades “Retos Investigación” y “Generación de Conocimiento” (Plan Estatal de Investigación Científica y Técnica y de Innovación 2017-2020). 136.972,00 EUR. 01/09/2021 - 31/08/2024
- 2010 - 2015 Grupo de Investigación del Sistema Universitario Vasco. “Análisis genómicos de la variación del ADN” (GIC10/58-IT558-10). Basque government
- 2007 - 2010 Plataforma web para gestión de conocimientos guiada por ontologías en genómica funcional. Proyecto Industrial- Regional Murcia (BIO-TEC 06/01-0005)

Transfer

- 2025 Cybersecurity Knowledge Graphs (2). Universidad de Murcia (UM), SIEMENS (14.000,00 EUR). Knowledge transfer project (TR42623) managed by Euskoiker foundation.
- 2024 - 2027 GAITECH. Company: LKS Next Gob Tech Euskadi; funding program: HAZITEK. Knowledge transfer project (PT10948) managed by Euskoiker foundation.
- 2024 Cybersecurity Knowledge Graphs. Universidad de Murcia (UM), SIEMENS (8.000,00 EUR). Knowledge transfer project (PT10885) managed by Euskoiker foundation.
- 2023 - 2024 Methods for automatic generation of Knowledge Graph. Universidad de Murcia (UM), BASF (8.000,00 EUR). Knowledge transfer project (TR42135) managed by Euskoiker foundation.
- 2021 - 2022 Producción de Datos Enlazados para Open Data Euskadi. EJIE (Sociedad Informática del Gobierno Vasco), 2105015. (9.000,00 EUR) Knowledge transfer project (TR41652) managed by Euskoiker foundation.

Development

- 2017 - 2018 SOLDAGE (Semantic Open Linked Data GEnerator). HAZITEK, Gobierno Vasco. FAIR data generator (150.000 EUR).
- 2016 - 2020 REPLICATE. Renaissance of Places with Innovative Citizenship And TEchnology (Project 691735), EU. Linked Open Data in Smart Cities (<http://replicate-project.eu/>). (Consortium: 24.965.263,09; Eurohelp: 328.580,00 EUR).
- 2015 - 2017 Linking Open Domains, Plataforma para la generación de datos enlazados (LODGen) (TSI-100105-2015-0012). Ministerio de Industria, Energía y Turismo (Spain), Acción Estratégica Economía y Sociedad Digital (AEESD) 1/2015. Linked Open Data pipeline (40.182,54 EUR).
- 2015 - 2016 Enlazando Gipuzkoa con el Mundo (ENGIMU). Gipuzkoako Foru Aldundia, Gipuzkoa IKT: Innovación digital Empresas (Spain). Linked Open Data pipeline (40.000 EUR).

Public tender

- 2018 Servicios Directorio. EJIE (Sociedad Informática del Gobierno Vasco). (70.000 EUR).
- 2016 - 2017 Servicios Open Linked Data. EJIE (Sociedad Informática del Gobierno Vasco). Linked Data implementation of Open Data Euskadi (90.000 EUR).

Publications

For publications please refer to <https://mikel-egana-aranguren.github.io/publications/>.

Committees and reviewer work

2025	Program Committee Member at Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA 2025)
2024	Reviewer for Journal of Biomedical Semantics (JBS)
2024	Reviewer for Bioinformatics (https://publons.com/wos-op/review/author/73gSS00l/)
2023	Program Committee Member at Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA 2023)
2022	Program Committee Member at Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA 2022)
2020	Program Committee Member at Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA 2020)
2019	Program Committee Member at Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA 2019)
2018	Program Committee Member at Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA 2018)

2017	Program Committee Member at Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWebMeDA 2017)
2017	Chapter review “Integrating Biological Data using Semantic Web Technology” in “Evolutionary Genomics. Computational and statistical methods”, 3rd edition, Springer
2015	Program Committee Member at Linked Data workshop (CAEPIA 2015)
2015	Reviewer for BMC Medical Informatics and Decision Making
2013	Special issue editor for Semantic Web Journal (SWJ): Special issue on Linked Data for Health Care and the Life Sciences
2013	Reviewer for PeerJ
2013	Reviewer for Data and Knowledge Engineering (DKE)
2012	Program Committee Member at Managing Interoperability and compleXity in Health Systems. In conjunction with the ACM International Conference on Information and Knowledge Management
2012	Program Committee Member at Joint Workshop on Semantic Technologies Applied to Biomedical Informatics and Individualized Medicine (SATBI + SWIM 2012). In conjunction with International Semantic Web Conference (ISWC)
2012	Reviewer for BMC Bioinformatics
2012	Reviewer for Journal of Biomedical Informatics (JBI)
2012	Reviewer for Computational and Mathematical Methods in Medicine (CMMM)
2012	Reviewer for Journal of Medical Systems (JOMS)
2012	Reviewer for Journal of Biomedical Semantics (JBS)
2011	Program Committee Member at Managing Interoperability and compleXity in Health Systems. In conjunction with the ACM International Conference on Information and Knowledge Management
2011	Program Committee Member at Knowledge Capture (K-CAP)
2011	Program Committee Member at Semantic Applied Technologies on Biomedical Informatics (SATBI 2011). In conjunction with the ACM International Conference on Bioinformatics and Computational Biology
2011	Reviewer for Semantic Web Journal (SWJ)
2011	Reviewer for Journal of Research and Practice in Information Technology (JRPIT)
2008	Program Committee Member at ONTORACT

Talks

2023	Grafos de conocimiento y transparencia. Bilbao Data Lab, Wikitoki, Euskal Herria
2016	Los Datos Enlazados y la Web Semántica. Tikitalka, VE Interactive, Spain
2014	Building reasonable biomedical ontologies for a Life Sciences Semantic Web. 3S (Systems, Synthetic, and Semantic) Biology summer school. CIBIO (Centre for Integrative Biology), University of Trento, Italy
2011	Linked Data for Functional Genomics. NTNU, Trondheim, Norway
2010	Aplicación de la Web Semántica en Biología Molecular. Universidad de Deusto, Facultad de Ingeniería, Spain

2008	Aplicación de la Web Semántica en Bioinformática. UM, Facultad de Informática, Spain
2004	Métodos y resultados actuales en Bioinformática: know-how y know-what de las redes tecnocientíficas en Bioinformática. EHU, Facultad de Filosofía, Spain

Teaching Experience

2025	Programazio Modularra eta Objektuetara Bideratutako Orientazioa (PMOBO). Degree in Computer Engineering of Management and Information Systems, EHU. Basque
2025	Administration of Data Bases (6 ECTS). Degree in Computer Engineering of Management and Information Systems, EHU. Spanish and Basque
2025	Administration of Data Bases (6 ECTS). Degree in Computer Engineering of Management and Information Systems, EHU. Spanish and Basque
2024	FAIR data (0.5 ECTS). MSc Bioinformatics, UM. Spanish
2024	Information Security (6 ECTS). Degree in Computer Engineering of Management and Information Systems, EHU. Spanish and Basque
2023	Administration of Data Bases (6 ECTS). Degree in Computer Engineering of Management and Information Systems, EHU. Spanish and Basque
2023	FAIR data (0.5 ECTS). MSc Bioinformatics, UM. Spanish
2023	Information Security (6 ECTS). Degree in Computer Engineering of Management and Information Systems, EHU. Spanish and Basque
2022	FAIR data (0.5 ECTS). MSc Bioinformatics, UM. Spanish
2022	Information Security (6 ECTS). Degree in Computer Engineering of Management and Information Systems, EHU. Spanish and Basque
2022	Analysis and Design of Information Systems (6 ECTS). Degree in Computer Engineering of Management and Information Systems, EHU. Spanish
2021	FAIR data (0.5 ECTS). MSc Bioinformatics, UM. Spanish
2021	Information Security (6 ECTS). Degree in Computer Engineering of Management and Information Systems, EHU. Spanish and Basque
2021	Analysis and Design of Information Systems (6 ECTS). Degree in Computer Engineering of Management and Information Systems, EHU. Spanish
2021	Project Management (3 ECTS). Degree in Computer Engineering of Management and Information Systems, EHU. Spanish and Basque
2020	FAIR data (0.5 ECTS). MSc Bioinformatics, UM. Spanish
2020	Information Security (6 ECTS). Degree in Computer Engineering of Management and Information Systems, EHU. Spanish
2020	Analysis and Design of Information Systems (6 ECTS). Degree in Computer Engineering of Management and Information Systems, EHU. Spanish
2017	Life Sciences Semantic Web (1 ECTS). MSc Bioinformatics, UM. Spanish
2016	Life Sciences Semantic Web (1 ECTS). MSc Bioinformatics, UM. Spanish
2016	Linked Open Data tutorial. EJIE. Spanish

2015	Linked Open Data tutorial. IZFE (Informatika Zerbitzuen Foru Elkartea, Gipuzkoa). Spanish
2015	Life Sciences Semantic Web (1 ECTS). MSc Bioinformatics, UM. Spanish
2014	Semantic biology tutorial: Use of Semantic Web resources for knowledge discovery. 3S (Systems, Synthetic, and Semantic) Biology summer school. CIBIO (Centre for Integrative Biology), University of Trento, Italy. English
2014	Galaxy tutorial. Erasmus mundus MSc in Marine Environment and resources, UPV-EHU. English
2014	Life Sciences Semantic Web (1 ECTS). MSc Bioinformatics, UM. Spanish
2013	Introductory talk on bioinformatics for high school students visiting the CBGP. Spanish
2013	Galaxy tutorials at CBGP. English and spanish
2013	Life Sciences Linked Data (1 ECTS). MSc Bioinformatics, UM. Spanish
2012	OWL, as part of ATHENS course (UPM). English
2012	OWL, as part of ATHENS course (UPM). English
2011	Populous tutorial at SWAT4LS (London, UK), English
2011	OWL, as part of ATHENS course (UPM). English
2011	Web Ontology Language (OWL), as part of Artificial Intelligence MSc (UPM). English
2011	OWL/Description Logics, as part of the Artificial Intelligence course (UPM). Spanish
2005-2008	OWL tutorials for biologists (University of Manchester, UK). English

Ph.D. panels

2016	Alejandro Rodríguez Iglesias, “FAIR approaches applied to unraveling plant-pathogen interactions data and RNA processing evolution”, UPM, Spain
2013	Meifania Monica Chen, “Lipoprotein Ontology: A Formal Representation of Lipoproteins”, Curtin University. Australia
2012	Jose Antonio Miñarro-Giménez, “Entorno para la gestión semántica de información biomédica en investigación traslacional”. UM, Spain
2011	Doris Mejía Ávila, “Estrategia de interoperabilidad semántica en el contexto de integración de conocimiento geográfico y ambiental. Caso de aplicación: Biodiversity Ontology”. UPM, Spain

Student supervision

2025	Bitor Delgado “Inyección de faltas en esquemas criptográficos post cuánticos”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
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2025	Iker Tranchand. “Cuántica aplicada a la detección de anomalías en empresas de ciberseguridad”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2025	Jon Izaguirre. “Evaluación de los espacios de datos en el contexto empresarial”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2025	Jon Marcos. “Evaluación empírica de frameworks MLOps”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2025	Bidane Leon. “IndusTroika: Plataforma de detección de anomalías para entornos industriales”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2025	Andreea Emilia Vasilica. “Garajeando: Aplicación Android para intercambio de coches entre vecinos”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2025	Hugo Robles. “Pokermón”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2024	Unai Elorriaga. “Research and evaluation of Openpilot through adversarial examples”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2024	Unai Castro. “App para la gestión de deportes entre amigos incluyendo estadísticas”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2024	Alberto Aróstegui. “Sistema ChatGPT-RAG para una mejor interacción con Jira”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2024	Alvaro Diez-Andino. “Cifrado asimétrico: Desde la perspectiva clásica a la cuántica a través de RSA”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2024	Mikel Perez. “Order superfast: App para la agilización de pedidos en restaurantes”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2024	Gorka del Río. “Aplicación móvil de intercambio de parejas con conocimiento cero”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2024	Igor Isasi. “Aplicación Web de exploración de datos en mapas basada en APIs REST de Open Data Euskadi”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2024	Paul Charrier. “Party Beast” Final Degree Project (TFG/GrAL), Erasmus+ (Ecole d’Ingenieurs ESME Sudria), EHU, Spain
2023	Joel García. “Aplicación de ChatGPT para la mejora de procesos de desarrollo en un entorno empresarial”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain

2023	Ena Verhorst. “Migración de una aplicación de gestión a una nube comercial usando integración continua”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2023	Aitor San Jose. “Desarrollo de una aplicación de soporte a revisiones médicas usando Odoo”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2023	Jon Garcia. “‘Técnicas para la gestión de secretos en entornos cloud’. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2023	Joseba Rodriguez. “Aplicación web basada en Nmap para monitorizar seguridad de servicios en una red corporativa”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2022	Maialen Ruiz. “Hezkuntzako ordezkapenak aukeratzeko mapa bidezko aplikazioa”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2022	Jon Ander Asua. “Grafo batean balizko ustelkeriaren datuak irudikatzeko framework generikoa”. Final Degree Project (TFG/GrAL), Bachelor’s Degree in Computer Engineering in Management and Information Systems, EHU, Spain
2015	Salvador Alonso Martínez. “Imagen Docker para pipelines de Metagenómica”, Bioinformatics MSc project (TFM), UM, Spain

Certifications

2023	Neo4J Certified Professional. Issued by Neo4J
2023	Semantic Integration Training. Issued by Semantic Web Company
2023	Knowledge Engineering Training. Issued by Semantic Web Company

Attended courses

2025	Aprende MongoDB desde cero. Udemy
2019	Docker Essentials: A Developer Introduction. Cognitive Class
2017	Predictive modelling. 20 hours course at Tecnalia
2003	Microsoft Visual Basic. 25 hours course at EHU
2003	Working risks prevention. 30h course at Bionatur Biologists Association (EHU)
2003	UNIX administration. 24 hours course at EHU
2002	ISO 9001:2000. 30h course at Bionatur Biologists Association (EHU)
2002	XML. 16 hours course at University of Deusto
1998	Environmental sciences (Ekoeskola). 65 hours course at Erreka Ecologist Platform

Technical expertise

Semantic Web stack: RDF (JSON-LD, TTL), SPARQL, OWL, SHACL, RML

Ontologies: SKOS, DCAT, VoID, Data Cube, WGS84, Open Biomedical Ontologies, Schema, European Legislation Identifier, and more

Ontology editors: Protégé

Databases: GraphDB, Stardog, Virtuoso, Blazegraph, RDFox, Neo4j, MongoDB, MariaDB

Linked/Open/FAIR Data: Trifid, Pool Party (Semantic Web Company), Corporate Memory (Eccenca GmbH), Morph-KGC, YARRRML, FAIR Data Point

Libraries: OWL API, RDFLib, Jena, RDF4J

Programming: Java, Python, Bash

Version control: Git (GitHub, GitLab)

Open Data: CKAN, Open Refine

Project management: Jira, Confluence, GitHub projects, Kanban

Cloud computing: Google Cloud, Azure

DevOps: GitHub actions, Docker, Docker compose, Ansible, Bamboo, Jenkins, Travis

Membership

Sociedad Española de Inteligencia Artificial en Biomedicina (BIOMED): <https://iabiomed.org/>

Sociedad Española de Bioinformática y Biología Computacional (SEBiBC): <https://sebirc.es/>